FACULTY OF COMPUTING

SEMESTER 1 2023/2024

SECP2613 SYSTEM ANALISIS AND DESIGN (WBL)

SECTION 1

ASSIGNMENT 1

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Question 1

(a) As the system analyst for ALLFOODS, I will evaluate the feasibility of Ahmad's proposed system project based on three aspects: technical feasibility, economic feasibility, and operational feasibility. During the discussion, Ahmad suggests a technical update, while Abu argues that the current system has no technical issues. Abu also points out that introducing a new system can be disruptive, and people generally resist change when the existing system is working well. Moreover, the significant cost of implementing a new system and the potential backlash from users could result in substantial losses for the company. In summary, the feasibility of this system project appears to be minimal.

(b) Operational feasibility is dependent on human resources available for the project and whether the system project able to operate and used by users if it is developed. Based on what Abu mentioned about, users satisfied about the current system as it meets the needs of users and may not see a pressing need for change. Moreover, the managers are also satisfied for the current system and there is not a must to having a newer system than Quality Foods. Therefore, the operational feasibility of the proposed project by Ahmad is low.

(c) Economic feasibility, also known as cost or benefit analysis, is utilized to assess whether the available time and funds are adequate for system development and to compare project costs and benefits. In a recent discussion, Abu emphasized that developing a new system product requires a substantial investment, and if users are dissatisfied with the product, the company could cause significant losses. On the other hand, Ahmad acknowledges the potential advantages of the new system but is unsure if these benefits will result in tangible financial profits. Given the risky nature of this venture, the economic feasibility of the proposed project is considered low.

(d) Technological feasibility refers to the technical requirements of the proposed project, assessing if the current technical resources are adequate for the new system. Ahmad aimed to enhance the system's technology, but Abu argued that their current system is functioning well and everyone is content with it. Additionally, Abu highlighted potential disruptions and integration challenges when introducing a new system into current operations. The high risk of success or failure in creating a new system must also be considered. Therefore, discussing the technological feasibility of the proposed project is crucial.

(e) After considering their conversation, I suggest conducting a comprehensive system study before moving forward with the proposed project. While Ahmad proposes an idea to impress users, Abu highlights potential disruptions and uncertainties regarding economic and operational feasibility. A detailed system study will analyse the current system, user requirements, and the costs and benefits of implementing a new system. This analysis will enable ALLFOODS to make a well-informed decision on whether investing in a new system is essential and advantageous for the organization.

Question 2

Activity		Predecessor	Duration
1	**Research Phase**		**9 weeks**
1.1	Define project scope and objectives	-	1 week
1.2	Interviewing with student and staff to gather requirements from them	1.1	1 week
1.3	Analysing system needs from existing registration system	-	2 weeks
1.4	Determining resource (Technical , Economic , Operational)	1.2 + 1.3	2 weeks
1.5	Make selection of project	1.4	2 weeks
1.6	Prepare and present system proposal	1.5	1 week
2	**Planning Phase**		**6 weeks**
2.1	Decide platform to develop	1.6	2 weeks
2.2	Develop a project plan timeline	2.1	1 week
2.3	Estimate project costs	2.1	2 weeks
2.4	Get approval for budget proposal	2.3	1 week
3	**Development Phase**		**17 weeks**
3.1	Design the user interface and human-computer interface	2.3	3 weeks
3.2	Design procedures for data entry	2.3	3 weeks
3.3	Develop backend system for student data management	3.2	4 weeks
3.4	Integrate necessary security measure	3.3	2 weeks
3.5	Design backup procedures	3.3	2 weeks
3.6	Implement the new system with programmers	3.5 + 3.4	2 weeks
3.7	Works with users to develop effective documentation	3.6	1 week
4	**Testing and Training Phase**		**2 weeks**
4.1	Test the information system	3.7	1 week
4.2	Do user training	4.1	1 week
5	**Evaluating and Maintenance Phase**		**Ongoing**
5.1	Roll out the new system	4.1	1 week
5.2	Monitor the system performance	5.1	Ongoing
5.3	Review and evaluate system	5.1	Ongoing

Question 3

In today's dynamic business environment, Agile Methodology has widely applied in various modern project management. There are several Agile Methodologies like Scrum, Extreme Programming and Kanban which are popular used enable a team to adapt with rapid changes and foster innovations. However, there are some differences will be faced by a manager in between traditional project management and this modern project management.

The first key difference is communication and collaboration between team members. The traditional project relies on heavily on face-to-face communication, formal meeting and formal documentation. But for Agile project, it focusses on in instant meeting, various collaborative tools and other informal interactions. This can enhance the information exchange and also improve the collaboration. Since the team members are from different time zone or locations, this method which focus on real-time communication can ensure the alignment between each member.

The second key difference is the project flexibility. The traditional project normally attributes fixed roles and responsibilities to each team member, and the project will follow the predefined project plans at most. But for Agile project, the plan will be always changed based on the continuous user feedback, the changing requirement. Moreover, each of the team members will be give different task to adapt with the changes in order to maximize the productivity and efficiency of the project. Therefore, this makes the team can always respond to the rapid changing environment.

The third key difference is the customer involvement. Traditional approaches only involve customer input primarily at the beginning to collect users' recommendation and at the end of the project to collect their feedbacks. In Agile project, the users will be encouraged to join thorough of the projects in order to get their real-time feedbacks. As the result, the team can identify issues early and fix them. This promotes a culture of continuous improvement and enhances the quality of result by addressing issues in a timely manner.

The fourth key difference is the risk management. Traditional project management always involves the risk assessment and back-up planning for a specific scope and timeline. But in Agile project, the managers will face more unknown changes and uncertainty than the traditional project, such as the miss understanding between each member. Therefore, the manager should construct more risk assessment, allow for frequent checkpoints and then adjust their backup plans accordingly in order to minimize the risks and make sure their project can cope with each uncertainty.

The last key difference is the way to handling change. Traditional methods always difficult to adapt with changes when the project is ongoing, and require re-planning process and potential delay. But for Agile project, they assume the changes and challenges are the natural part of the project. By continuously collecting feedback and making adjustment, the team enable to respond quickly to changing priorities and market conditions and reduce the risk of project failure due to uncertainty.

In short, that is important for all business projects in nowadays business environment shift from traditional to Agile method, not only to improve the enhance the team's ability to adapt with the rapid changes of the environment, but also improve the overall productivity and efficiency of the team. As the result, using Agile method enable a distributed team to stand in the complex and fast-changing business environment.

Question 4

a)

ESTIMATED COST	RM (RINGGIT)	
Hardware	50000	
Software	15000	
Training	30000	
Consulting	80000	
Data Conversion	40000	
Supplies	15000	per year
IS Salaries	75000	in year 1; 10% annual increase
Upgrades	10000	per year over 3 years

ASSUMPTIONS	Percentage (%)
Discount rate	35%

ESTIMATED BENEFITS	RM (RINGGIT)	
Improve Customer Service	150000	in year 1; 20% increase anually
Increase productivity	150000	per year; 25% increase anually

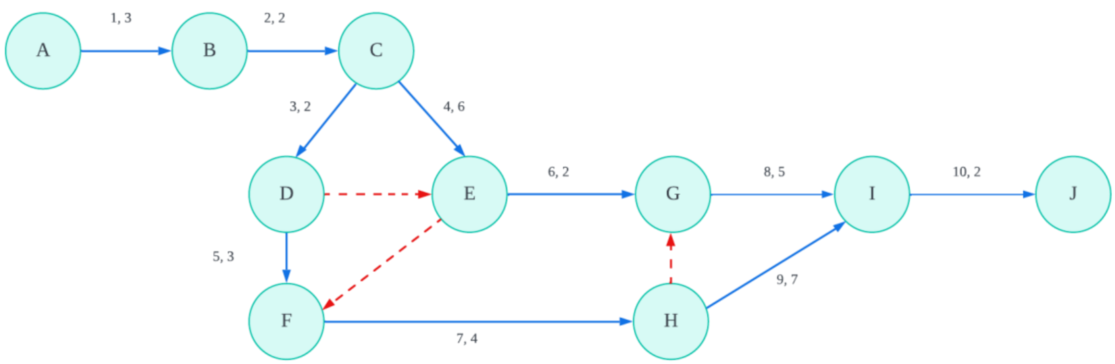
Costs	Year 0	Year 1	Year 2	Year 3
Development Costs				
Hardware	50 000			
Software	15 000			
Training	30 000			
Consulting	80 000			
Data Conversion	40 000			
Total	**215 000**			
Production Costs				
Supplies		15 000	15000	15000
IS Salaries		75 000	82 500	90 750
Upgrades		10 000	10 000	10 000
Annual Production Costs (Total)		**100 000**	**107 500**	**115 750**
Annual Production Costs (Present Value)		**74 074**	**58985**	**47046**
Accumulated Costs		**289 074**	**348 059**	**395 105**

Benefits	Year 0	Year 1	Year 2	Year 3
Improve Customer Service		150 000	180 000	216 000
Increase productivity		150 000	187 500	234 375
Total		**300 000**	**367 500**	**450 375**
Present Value		222 222	201646	183 051
Acumulated Benefits (Present Value)		**222 222**	**423 868**	**606 919**
Gain or Loss		**(66 852)**	**75 809**	**211 814**
Profitability Index	**0.99**			

b) The probability index (PI) for this Present Value (PV) analysis of SKORBISTARI is 0.99 which showing it is not an excellent investment as its index is less than one.

Question 5

a) PERT diagram



b) All paths

Path 1: 1-2-3-6-8-10
Path 2: 1-2-3-5-7-9-10
Path 3: 1-2-3-5-7-8-10
Path 4: 1-2-4-7-8-10
Path 5: 1-2-4-6-8-10
Path 6: 1-2-4-7-9-10

c) Critical path

Path 1 → Length=3+2+2+2+5+2= 16 weeks
Path 2 → Length=3+2+2+3+4+7+2= 23 weeks
Path 3 → Length=3+2+2+3+4+5+2= 21 weeks
Path 4 → Length=3+2+6+4+5+2= 22 weeks
Path 5 → Length=3+2+6+2+5+2= 20 weeks
Path 6 → Length=3+2+6+4+7+2= **24 weeks**

Therefore, the critical path, which is the longest path, is Path 6, 1-2-4-7-9-10.

d) Gantt Chart